Extra Bite #1: Power Supplies and Regulators

Clean DC, in plain English. One pass over E7D's 15 questions.

Group E7D · 15 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Why You Care

Every radio runs on DC. The wall gives us AC. Something has to turn AC into clean, steady DC and hold the voltage where the radio expects it – even when the load changes or the line voltage sags. That something is the power supply, and inside it is a **regulator**. Extra asks you to recognize the two families, name a few standard parts, and do one easy power calculation.

The Two Families

Type	How it holds the voltage	Trade-off
Linear	A pass transistor in series with the load. The circuit smoothly varies how hard that transistor conducts to keep the output constant. (E7D01)	Simple, quiet, burns the difference as heat. Heavy.
Switching	The input is chopped into high-frequency pulses, and the regulator varies the **duty cycle** of those pulses into a filter. The filter averages the pulses back to steady DC. (E7D02)	Lighter, cheaper, more efficient. High frequency shrinks the transformer and filter parts. (E7D10)

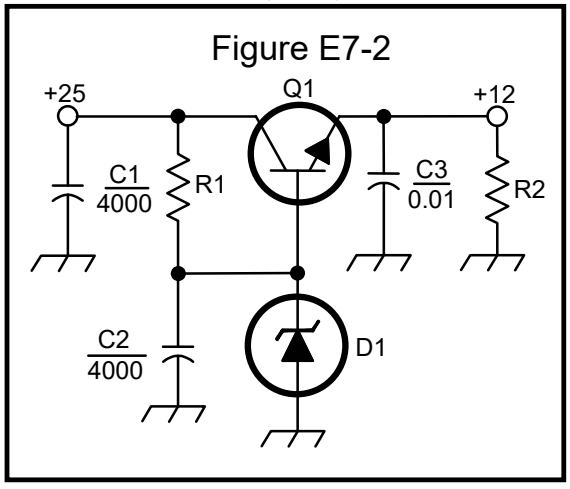
Two flavors of linear regulator

- **Series** regulator: control element is in line with the load. A **three-terminal regulator** (78xx, LM317) is a series regulator in one chip. (E7D04)
- **Shunt** regulator: control element is across the load and pulls extra current from the unregulated source to hold the output down. So it **loads the source even at no load**. (E7D05)

Reading Figure E7-2 (Official NCVEC Diagram)

This is the actual diagram you will see on the exam. Questions E7D06, E7D07, and E7D08 refer to **this** figure.

Figure E7-2



Orientation: +25 V comes in top-left; +12 V regulated comes out top-right. The output passes through Q1 (NPN pass transistor) on its way to the load. Bottom rail is ground.

- **D1 (Zener diode)** = the voltage reference. Holds a known voltage that the regulator compares against.
- **Q1 (pass transistor)** = the control element. **Q1 controls the current to keep the output voltage constant.** (E7D06)
- **C2** bypasses leftover rectifier ripple around D1, keeping the reference clean. (E7D07)
- C1, C3 are filter/bypass caps; R1, R2 are bias resistors (not exam-tested).
- Whole circuit type: **linear voltage regulator**. (E7D08)

Three Numbers You Will Be Asked

Concept	Formula or rule	Example
Dropout voltage (E7D12)	Minimum input-to-output gap that still gives proper regulation. Below it, the output sags with the input.	A 12 V linear regulator with 2 V dropout needs at least 14 V in.
Series-regulator power dissipation (E7D13)	$P = (V_{in} - V_{out}) \times I_{out}$. The pass transistor burns the voltage difference at the load current.	18 V in, 12 V out, 2 A load: P = (18−12) × 2 = 12 W as heat.
Battery operating time (E7D09)	Time (hours) = Capacity (Ah) ÷ Average current (A).	20 Ah battery at 4 A average: 20 / 4 = 5 hours.

The Other Five Facts

Topic	What to remember	Q
Voltage reference	A **Zener diode** is the standard stable voltage reference inside a regulator.	E7D03
Solar panel inverter	A solar panel produces **DC**. An inverter on its output converts DC to AC.	E7D11
Bleeder resistors on stacked filter caps	Equal resistors across each cap do three jobs: equalize the voltage across each cap, discharge them when off, and provide a minimum load. (Answer: all three.)	E7D14

Step-start circuit	Inrush limiter that lets the filter caps **charge gradually** at switch-on rather than taking a large current spike through the rectifier.	E7D15

Mental Cheat Sheet

- Linear = pass transistor varies its conduction. Heavy, hot, clean.
- Switching = pulses + filter, duty cycle varies. Light, cool, slightly noisy.
- Three-terminal regulator = series linear in a chip.
- Shunt regulator loads the source even at no load.
- Voltage reference = Zener diode.
- Q1 in Figure E7-2 = pass transistor controlling output current.
- Dropout = minimum ($V_{in} - V_{out}$) for regulation.
- Series-regulator heat = ($V_{in} - V_{out}$) × I_{out}.
- Battery time = Ah ÷ current.
- Inverter on solar = DC to AC.
- Equal Rs on stacked filter caps = balance + bleed + min load (all three).
- Step-start = gradual charge of filter caps.

Common Traps

- The **shunt** regulator loads the source – not the series. Students reflexively pick “series.”
- Switching supplies vary **duty cycle** of pulses. “Varying the conductivity of a pass element” is the linear answer.
- An inverter converts **DC to AC**. Not a filter, not a regulator, not a battery.
- The bleeder-resistor question has three legitimate functions, so the answer is “all of these.”

Takeaway: A power supply turns AC into DC; a regulator holds the DC steady either by burning the difference as heat in a series transistor (linear) or by chopping and filtering pulses (switching). Five extra facts plus one easy power formula round it out.

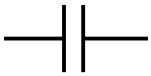
Electronics Reminders From The General Exam Material

If the schematic symbols look like alphabet soup, read this first. Most of it was on Technician and General, but most of us pattern-matched our way through. This is your second pass – useful for every Bite, not just this one.

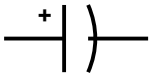
The Common Symbols



Resistor — label R, units ohms (Ω, k, M)
Limits current and drops voltage. Zigzag is US style; you may also see a plain rectangle (international).



Capacitor (non-polar) — label C, units farads (pF, nF, µF)
Stores charge on two plates. Blocks DC; lets AC pass.



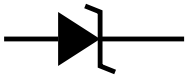
Capacitor (polarized) — label C
Same job, but must connect (+) to the more-positive voltage or it fails loudly. Used for big values in power supplies.



Inductor — label L, units henries (µH, mH, H)
A coil. Stores energy in a magnetic field. Passes DC; blocks high-frequency AC. Often paired with caps to make filters.



Diode — label D
A one-way valve for current. Flows the direction the triangle points; blocked the other way. Bar at tip = cathode.



Zener diode — label D
A diode designed to "break down" at a precise reverse voltage and hold there. Used as the stable voltage reference (D1 in Figure E7-2).



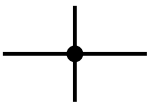
NPN transistor — label Q, 3 pins (B, C, E)
A controlled current valve. Small current into the Base controls a larger Collector-to-Emitter current. Arrow on Emitter points **out**. Q1 in E7-2 is NPN.



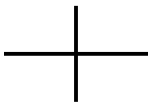
PNP transistor — label Q
Reverse of NPN. Arrow on Emitter points **in** toward the Base; current flows from Emitter to Collector. Used in the upper half of complementary stages.



Ground — label GND
The 0-volt reference for the whole circuit. All ground symbols on a page connect to the same point electrically, even with no drawn wire.



Junction (dot)
A filled black dot at a crossing means the wires are electrically connected.



Crossover (no dot)
No dot at a crossing means the wires pass each other and are **not** connected. This trips up beginners constantly.

Rule of thumb
Always check for the dot at a crossing. The presence or absence of one black dot completely changes how the circuit works.

How To Read Component Values

What you see	What it means
4.7k on a resistor	4,700 ohms ("k" = times 1,000)
4000 on a resistor	4,000 ohms (no multiplier letter = plain ohms)
1M on a resistor	1,000,000 ohms (capital M = mega = times 1,000,000)
10µF or **10uF** on a capacitor	10 microfarads (millionths of a farad)
100pF on a capacitor	100 picofarads (millionths of a microfarad)
0.01 on a capacitor (no unit, just a number)	0.01 microfarads, by long-standing convention. A bare small decimal on a cap is µF.
10mH on an inductor	10 millihenries (thousandths of a henry)

Electronics Reminders (continued)

Multiplier Prefixes

Prefix	Letter	Multiplier	In words
pico	p	0.000 000 000 001	one trillionth
nano	n	0.000 000 001	one billionth
micro	µ or u	0.000 001	one millionth
milli	m	0.001	one thousandth
(none)	—	1	the base unit
kilo	k	1,000	one thousand
mega	M	1,000,000	one million
giga	G	1,000,000,000	one billion

Reference Designators (Letters Before The Numbers)

Every component on a schematic has a letter-and-number tag. The letter tells you the type.

Letter	Component type	Letter	Component type
R	Resistor (R1, R2)	**Q**	Transistor (Q1)
C	Capacitor (C1, C2, C3)	**U**	Integrated circuit (U1)
L	Inductor (L1)	**T**	Transformer (T1)
D (or CR)	Diode, including Zener (D1)	**J**	Jack / connector (J1)

Power Rails and Voltage Labels

You will often see labels like **+25**, **+12**, **+5**, or **−12** on a schematic. These mark supply rails. A “+12” anywhere on the page means that point connects to the +12 V supply, even with no wire drawn there. Ground is the implied 0 V rail. Voltages are always measured between two points; in practice, between a point and ground.

A Habit That Helps

Before reading the prose around any schematic, take 30 seconds:

- Find the input (usually left, often a positive voltage like +25).
- Find the output (usually right, the regulated voltage like +12).
- Find ground (bottom rail or any ground symbol).
- Trace the path the current takes from input to output.
- Then read the labels (Q1, D1, etc.) and ask “what is this part doing on that path?”

This works for almost every Extra schematic. The questions almost always ask “what is part X doing?” Once you can name the path and the parts along it, the answers pick themselves.

Bite 1 Question Bank (E7D)

15 questions. Read, decide, check the answer key at the end.

E7D01. How does a linear electronic voltage regulator work?
A. It has a ramp voltage as its output.
B. It eliminates the need for a pass transistor.
C. The control element duty cycle is proportional to the line or load conditions.
D. The conduction of a control element is varied to maintain a constant output voltage.

E7D02. How does a switchmode voltage regulator work?
A. By alternating the output between positive and negative voltages.
B. By varying the duty cycle of pulses input to a filter.
C. By varying the conductivity of a pass element.
D. By switching between two Zener diode reference voltages.

E7D03. What device is used as a stable voltage reference?
A. A Zener diode.
B. A digital-to-analog converter.
C. An SCR.
D. An analog-to-digital converter.

E7D04. Which of the following describes a three-terminal voltage regulator?
A. A series current source.
B. A series regulator.
C. A shunt regulator.
D. A shunt current source.

E7D05. Which of the following types of linear voltage regulator operates by loading the unregulated voltage source?
A. A constant current source.
B. A series regulator.
C. A shunt current source.
D. A shunt regulator.

E7D06. What is the purpose of Q1 in the circuit shown in Figure E7-2?
A. It provides negative feedback to improve regulation.
B. It provides a constant load for the voltage source.
C. It controls the current to keep the output voltage constant.
D. It provides regulation by switching or chopping the input DC voltage.

E7D07. What is the purpose of C2 in the circuit shown in Figure E7-2?
A. It bypasses rectifier output ripple around D1.
B. It is a brute force filter for the output.
C. To prevent self-oscillation.
D. To provide fixed DC bias for Q1.

E7D08. What type of circuit is shown in Figure E7-2?
A. Switching voltage regulator.
B. Common emitter amplifier.
C. Linear voltage regulator.
D. Common base amplifier.

E7D09. How is battery operating time calculated?
A. Average current divided by capacity in amp-hours.
B. Average current divided by internal resistance.
C. Capacity in amp-hours divided by average current.
D. Internal resistance divided by average current.

E7D10. Why is a switching type power supply less expensive and lighter than an equivalent linear power supply?
A. The inverter design does not require an output filter circuit.
B. The control circuitry uses less current, therefore smaller heat sinks are required.
C. The high frequency inverter design uses much smaller transformers and filter components for an equivalent power output.
D. It recovers power from the unused portion of the AC cycle, thus using fewer components.

E7D11. What is the purpose of an inverter connected to a solar panel output?
A. Reduce AC ripple on the output.
B. Maintain voltage with varying illumination levels.
C. Prevent discharge when panel is not illuminated.
D. Convert the panel's output from DC to AC.

E7D12. What is the dropout voltage of a linear voltage regulator?
A. Minimum input voltage for rated power dissipation.
B. Maximum output voltage drop when the input voltage is varied over its specified range.
C. Minimum input-to-output voltage required to maintain regulation.
D. Maximum that the output voltage may decrease at rated load.

E7D13. Which of the following calculates power dissipated by a series linear voltage regulator?
A. Input voltage multiplied by input current.
B. Input voltage divided by output current.
C. Voltage difference from input to output multiplied by output current.
D. Output voltage multiplied by output current.

E7D14. What is the purpose of connecting equal-value resistors across power supply filter capacitors connected in series?
A. Equalize the voltage across each capacitor.
B. Discharge the capacitors when voltage is removed.
C. Provide a minimum load on the supply.
D. All these choices are correct.

E7D15. What is the purpose of a step-start circuit in a high-voltage power supply?
A. To provide a dual-voltage output for reduced power applications.
B. To compensate for variations of the incoming line voltage.
C. To prevent arcing across the input power switch or relay contacts.
D. To allow the filter capacitors to charge gradually.

Answer Key

E7D01: D	E7D02: B	E7D03: A
E7D04: B	E7D05: D	E7D06: C
E7D07: A	E7D08: C	E7D09: C
E7D10: C	E7D11: D	E7D12: C
E7D13: C	E7D14: D	E7D15: D